

Identifying Plant Remains

Identifying Seeds

Seeds are identified through morphological comparison with modern seeds from known plants. This allows a paleoethnobotanist to compare the size, shape, and surface features (e.g., texture, furrows, or other identifying marks) of modern and ancient seeds to make a match. This approach has been greatly facilitated by modern seed-image databases online (e.g., the [Digital Plant Atlas](https://www.plantatlas.eu/) (<https://www.plantatlas.eu/>)), but many laboratories, including the one at Boston University, have their own physical collections of plant parts and seeds that are used to identify archaeological specimens.

Explore images from the BU Environmental Archaeology Lab's [Flickr collections](https://www.flickr.com/photos/104202247@N08/albums/72157648282139182/) (<https://www.flickr.com/photos/104202247@N08/albums/72157648282139182/>) and [full database](https://sites.bu.edu/ealab/collections/database/) (<https://sites.bu.edu/ealab/collections/database/>).

Identifying Phytoliths

Phytoliths are also identified through morphological comparison, although their complex three-dimensional shapes can make this more difficult. It is necessary to rotate many phytoliths during identification, which is a challenge, as they are microscopic and float in a solution on a glass slide. This rotation can be accomplished by tapping the slide with a pencil eraser or another soft, pointed object, which will move the liquid and roll phytoliths around, though often randomly. Phytoliths are often compared with reference images, as these images can be taken from multiple angles to show the three-dimensional form of a phytolith type, and with illustrations.

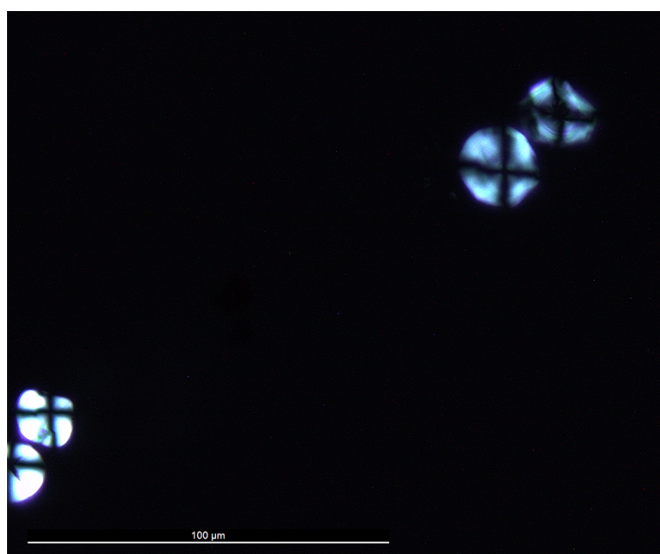
The greater challenge with phytoliths is that specific phytolith forms may be shared among a broad or narrow lineage of plants, ranging from an entire family containing hundreds of species to a single tissue (e.g., the leaf) of a single species of plant. Phytolith analysts are trained to learn the differences between phytoliths identifiable at the general level and those that lead to more specific identifications.

Identifying Pollen

Pollen is also identified morphologically, using the size, shape, and surface texture of the pollen grain as the primary tools. Reference collections of modern pollen are helpful resources, as are images, those published in both books and increasingly online reference collections. One of the best for identifying pollen from archaeological sites is the UMass Boston Human Impacts Pollen Collection (<https://keyserver.lucidcentral.org/key-server/data/0f030b07-0200-4b0f-8509-0a1808956702/media/Htm/index.html>).

Identifying Starch

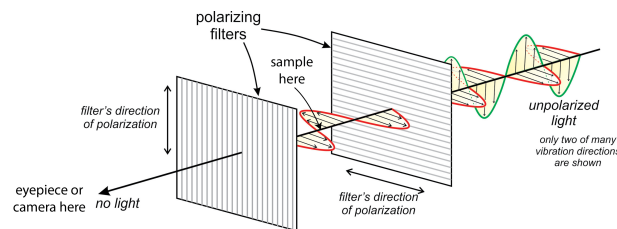
Starch is identified through morphological comparison with modern reference collections, either physical collections on slides or image collections, either digital or in published resources. Importantly, however, due to the **birefringence** properties of starch, it is critical to examine starch granules through both ordinary and **cross-polarized** light. Birefringence in the starch granule produces a feature termed the “extinction cross” under cross-polarized light, as shown in the images below. Features of the extinction cross differ among different types of starch: The shape, location of the center, and position of the “arms” of the cross vary.



Maize starch granule, under ordinary transmitted light (above) and cross-polarized light (below). Note that the extinction cross in maize is centered, or close to it, on the relatively round starch granule.

Source: Dr. Marston

Cross-polarized light involves using two filters. The first ensures that only specific, parallel wavelengths of light pass through the sample (initial polarization). That light is transmitted through the sample and passes through another polarizing filter oriented perpendicularly to the original filter. This blocks all light that has not been bent into another plane by passing through the sample; this is why the background color changes from light to dark. Only the samples with refringent properties that have changed the plane of transmitted light appear in the image.



Source: Modified from OpenGeology's [Introduction to Mineralogy](https://opengeology.org/Mineralogy/5-optical-mineralogy/) (<https://opengeology.org/Mineralogy/5-optical-mineralogy/>), Ch. 5